

Euclid's Elements — Book V

Proposition 10

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

A strict comparison of ratios with a common term reflects the corresponding strict comparison of magnitudes.

1 Introduction

Proposition V.10 is the order-reflection companion to Proposition V.8. Where V.8 starts with a strict inequality of magnitudes and constructs a strict inequality of ratios, V.10 starts with a strict ratio comparison and recovers the underlying strict inequality of magnitudes.

In the present reconstruction this converse direction is substantially shorter than V.8. No Archimedean argument and no least-exceeding-multiple construction are needed. The proof simply unpacks the witness supplied by Eudoxus' greater-ratio relation, applies trichotomy to the candidate magnitudes, and excludes the two wrong comparison cases.

The production file is

```
CGJteamLab/Proposition5_10.lean
```

and it imports only

```
CGJteamLab.HilbertInterfaceV
```

from the project-level Book V infrastructure.

2 Formal Reading of Proposition V.10

The production file packages two comparison forms.

The first says:

$$a : d > c : d \implies c < a.$$

The ratios have the same second term d .

The second says:

$$a : b > a : d \implies b < d.$$

The ratios have the same first term a .

These are exactly the two forms needed later in the reconstruction of V.14.

3 The Greater-Ratio Witness

Recall that

HilbertEudoxusGreaterRatio Geo a b c d

means that there exist positive-multiple indices m, n such that

$$M_n(b) < M_m(a)$$

and

$$\neg(M_n(d) < M_m(c)).$$

Thus a strict comparison of ratios is not represented by a scalar quotient. It is represented by a concrete pair of finite multiples that separates the two ratios.

V.10 shows that when one term is shared, such a witness forces the expected order of the remaining magnitudes.

4 First Comparison Form

Assume

$$a : d > c : d.$$

Unpacking the greater-ratio witness gives m, n with

$$M_n(d) < M_m(a)$$

and

$$\neg(M_n(d) < M_m(c)).$$

Now apply trichotomy to c and a .

There are three cases:

$$c < a, \quad c = a, \quad a < c.$$

The first case is the desired conclusion.

If

$$c = a,$$

then the two multiple comparisons in the witness become the same proposition. The witness would simultaneously assert

$$M_n(d) < M_m(a)$$

and its negation. This is impossible.

If instead

$$a < c,$$

then positive multiplication preserves strict order:

$$M_m(a) < M_m(c).$$

Together with

$$M_n(d) < M_m(a),$$

transitivity yields

$$M_n(d) < M_m(c),$$

contradicting the negative clause of the greater-ratio witness.

Therefore only

$$c < a$$

remains.

5 Second Comparison Form

Assume

$$a : b > a : d.$$

The witness gives

$$M_n(b) < M_m(a)$$

and

$$\neg(M_n(d) < M_m(a)).$$

Apply trichotomy to b and d .

If

$$b < d,$$

we are done.

If

$$b = d,$$

the positive and negative witness clauses become contradictory.

If

$$d < b,$$

positive multiplication gives

$$M_n(d) < M_n(b).$$

Combining this with

$$M_n(b) < M_m(a)$$

gives

$$M_n(d) < M_m(a),$$

again contradicting the negative witness clause.

Hence

$$b < d.$$

6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_5_10
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (a b c d : HilbertPositiveSegmentClass Geo) :
  (HilbertEudoxusGreaterRatio Geo a d c d ->
    HilbertPositiveSegmentLess Geo c a) /\
  (HilbertEudoxusGreaterRatio Geo a b a d ->
    HilbertPositiveSegmentLess Geo b d) := by

constructor
case left =>
  exact
    hilbertEudoxus_v10_same_second_lt_first
      Geo a c d
case right =>
  exact
    hilbertEudoxus_v10_same_first_lt_second
      Geo a b d

```

As in V.8, the final Euclid-numbered theorem is a thin wrapper over two proposition-local lemmas:

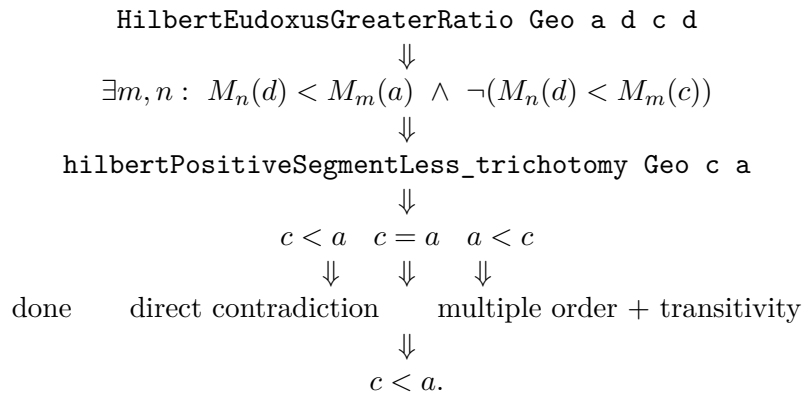
```

hilbertEudoxus_v10_same_second_lt_first
hilbertEudoxus_v10_same_first_lt_second

```

7 Lean Proof Architecture

The first half has the following formal DAG:



The second half is identical in structure, with trichotomy applied to b, d .

8 Infrastructure Used

Unlike V.8, the proof does not use

```
HilbertArchimedeanPlane  
hilbertPositiveSegment_archimedean  
hilbertPositiveSegment_exists_least_exceeding_multiple
```

at all.

The essential reusable infrastructure is only:

```
hilbertPositiveSegmentLess_trichotomy  
hilbertPositiveSegmentMultiple_lt  
hilbertPositiveSegmentLess_trans
```

together with the definition of

```
HilbertEudoxusGreaterRatio
```

itself.

This is mathematically significant: V.10 is an order-theoretic consequence of the Eudoxean witness once the existence of that witness is assumed.

9 Relationship with V.8

V.8 and V.10 form a natural pair.

V.8 gives, in the common-second-term form,

$$c < a \implies a : d > c : d.$$

V.10 gives the converse:

$$a : d > c : d \implies c < a.$$

Similarly, for a common first term,

$$b < d \implies a : b > a : d$$

comes from V.8, while

$$a : b > a : d \implies b < d$$

comes from V.10.

Thus the two propositions together identify strict order of magnitudes with the corresponding strict ratio comparison in these common-term configurations.

10 Role in Proposition V.14

V.10 is used later as the final step in the strict comparison part of V.14.

Suppose

$$a : b = c : d$$

and

$$a < c.$$

V.8 first gives a strict ratio comparison with a common second term. V.13 transports that comparison through the equality of ratios. V.10 then converts the resulting ratio comparison back into

$$b < d.$$

The dependency chain is therefore

$$\text{V.8} \longrightarrow \text{V.13} \longrightarrow \text{V.10} \longrightarrow \text{V.14}.$$

11 Hidden Formal Data

Two features of the Lean proof deserve attention.

11.1 The witness indices are retained

The proof does not discard the particular m, n supplied by the greater-ratio relation. Those same indices are used to derive the contradiction in the wrong trichotomy case.

11.2 Order preservation under a fixed positive multiple

The crucial move in the wrong strict-order case is

$$a < c \implies M_m(a) < M_m(c),$$

or, in the second half,

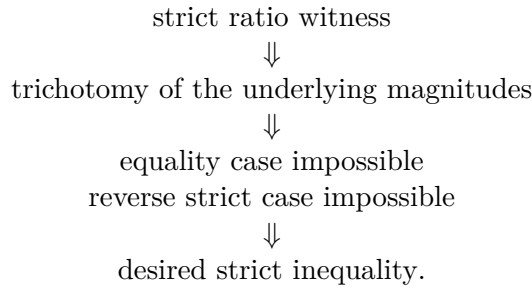
$$d < b \implies M_n(d) < M_n(b).$$

This is the role of

and explains why the positive-multiple order theory belongs in the reusable Book V interface rather than in a proposition-specific file.

12 Classical DAG versus Formal DAG

The conceptual proof is extremely short:



The Lean proof follows exactly this structure.

There is no auxiliary geometric construction and no proposition-specific magnitude object. The extra formal detail consists only in spelling out the multiple indices and the transitivity step that produces the contradiction.

13 Structural Lesson

V.10 separates two logically distinct layers of the Book V development.

The difficult existence problem belongs to V.8: one must construct suitable multiple indices witnessing a strict ratio comparison.

Once such indices exist, V.10 is almost purely order logic. It demonstrates that the Eudoxean Definition 7 is strong enough to reflect strict magnitude order in the presence of a common ratio term.

This division is useful for the later architecture. V.14 does not need to repeat any Archimedean reasoning: it uses V.8 to manufacture a ratio comparison, V.13 to transport it, and V.10 to read the resulting comparison back as a magnitude inequality.

14 Conclusion

Proposition V.10 is the order-reflection theorem of the reconstructed Eudoxean ratio calculus.

Its proof is short because all existence content has already been packaged in the definition of a greater ratio. By trichotomy, preservation of strict order under positive multiplication, and transitivity, the two wrong magnitude comparisons are excluded.

The result is the exact converse mechanism needed after V.8 and is a central input to V.14.